

Guessing the Answer

Goals For Today

- Guess the form of an answer before starting using dimensional analysis
- Use dimensional analysis and limiting cases to check your answers
- Construct back-of-the-envelope answers without complex math/physics
- Continue building the framework for approaching a problem

Dimensional Analysis¹

Dimensional analysis is a strategy that involves attaching **dimensions** or **units** to the variables and values in your equations, and keeping track of these dimensions throughout your calculations. (Units are things like meters, feet, or centimeters; dimensions are things like length. Using the latter is nice because our units may change while referring to the same fundamental type of quantity.) Dimensional analysis can be used in two contexts: the first is after obtaining an answer, checking that the dimensions match the those of the quantity you're looking for. Let's do a quick example.

Kicking off.² Say you kick a soccer ball at speed v and angle θ from the horizontal. We'll assume constant gravity, g , and ignore air resistance. Say we want to know the horizontal distance d that the ball goes before hitting the ground.

1. Might the formula $d = 2v^2 \cos \theta$ yield a reasonable answer? Why or why not?

2. How about $d = 2\frac{v^2}{g} \sin g$?

3. How about $d = 2\frac{v^2}{g} \sin \theta$?

¹I highly recommend [these notes](#) for a more detailed explanation and systematic approach to dimensional analysis.

²This problem is adapted from [these notes](#).

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4. Another sanity check: thinking about dimensions, what do e^x , $\ln x$ and $\sin x$ have in common? How can we use this to check a result? List more members of this family.

The second way we use dimensional analysis is to “guess” the form of a solution before we even begin. This will allow you to form a rough solution to work *towards* as you calculate.

Troy’s Problem. Let’s say we have a pendulum, of length ℓ , which has a mass m attached at its end. The only force acting on the pendulum is Earth’s gravity, g , which is uniform. What is the period of the pendulum’s motion, T ?

1. What dimension does T have?
2. What dimensions do ℓ, m, g each have?
3. What would you guess the formula for T to be based on your answers to questions one and two? (Hint: you don’t need to use all the variables!)
4. What kind of values can appear in equations that are dimensionless?³ What might that lead you to conclude about your formula?

Limiting Cases

Limiting cases allow you to check your answer by testing it in a scenario you already know the answer to. Just like dimensional analysis, limiting cases can be used to check your work or to develop a guess for the solution. Generally, the cases you know the answer to are “extreme” in some way - for example, you can compare your intuitive, physical knowledge of how a system behaves when plugging in values like 0 or ∞ , to the mathematical answer you

³Technically, these kinds of quantities are ‘of dimension one.’

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calculate using your own expression.

Return of the Soccer Player. We saw in the previous section that $d = 2\frac{v^2}{g} \sin \theta$ has consistent dimensions. But what are some other ways we can check our result?

1. In what situation(s) would the ball travel zero horizontal distance? Does our equation yield a result in these cases that makes physical sense? (*Hint:* there are multiple answers.)
2. How might you modify the original equation to yield consistent results with the scenario(s) you explored above?
3. What result do you expect for $\theta = 0$? Remember that you are solving for the horizontal distance *before the ball hits the ground*. Is this result consistent with what you calculate using your modified equation? If not, how should you adjust your equation?
4. Is there a situation in which the ball could travel negative horizontal distance? Test your modified equation above in the case that the ball goes backwards. Does it work?
5. (Optional.) Use standard kinematic equations to confirm your answer.

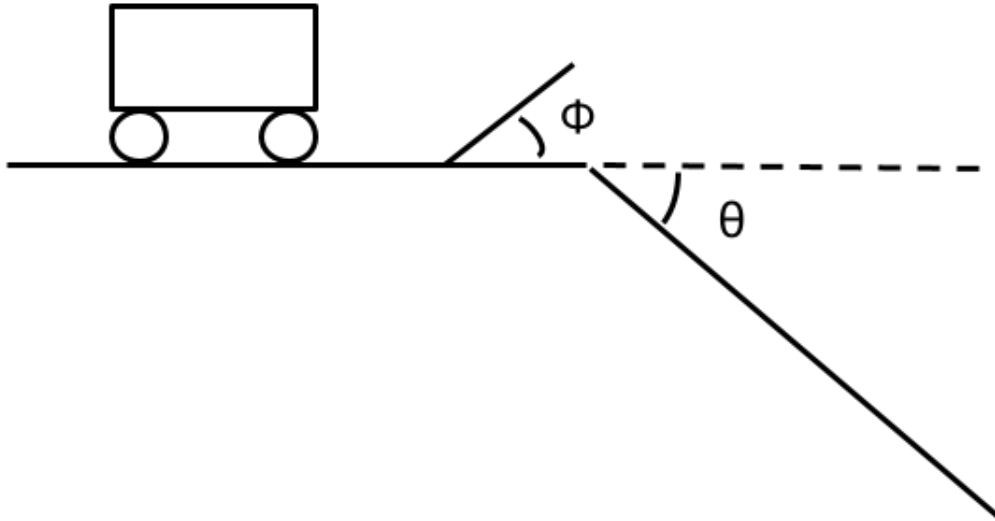
Try using the limiting cases method on the example problem below.

Fast and Furious

You are in a car chase and need to gain as much ground on your pursuers as you can. One of your friends is nearby and is setting up a ramp at an angle θ which you can select. At the

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end of a ramp is a hill that slopes uniformly downward at angle ϕ . At what angle θ should your friend set up the ramp so that you jump the furthest?



1. What are we searching for in this problem? What might the form of the answer look like? (i.e., if it's a function, will it be linear/quadratic/something else? If it's a value, what are the constraints on it?)
2. What are a few 'extreme'/'limiting' cases that you can guess the answers to? (Hint: good extreme cases come from the 'edge' values/constraints on the problem.)
3. Based on your answers to the previous three questions, 'guess' an answer to this problem.

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4. What assumptions did you make in coming up with your guess? Do you think these are reasonable?

5. (Optional.) Use standard kinematics equations to confirm your answer to question 4. (Hint: check example 1.10 of K&K.)

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Appendix: What does that “equals” sign mean?

When approximating, it is useful to describe different types of relationships values can have with each other. Below are some you might see. Note that the definitions I’ve given here are how you will probably see them in physics classes but are not necessarily the right definitions in math.

$a \equiv b$ or $a := b$: a is defined to be the same as b . This means that a and b don’t just have the same value, they represent the same thing at a deeper level. For example, if a circle has radius r , its diameter d has $d \equiv 2r$. a and b have the same units.

$a = b$: a and b have the same value. For instance, $2 + 4 = 5 + 1$. a and b have the same units.

$a \simeq b$ or $a \approx b$: a and b are pretty close to equal to each other. For example, $\pi \approx 3.14$. What “pretty close” means is context dependent. a and b have the same units.

$a \propto b$: a is proportional to b , meaning that $a = Cb$ for some constant value C . a and b don’t necessarily have the same units, because C is allowed to have units.

$a \sim b$: a “scales” the same way as b , meaning that even though they might not have the same value (or even be close to each other), a should be roughly proportional to b , proportional to b in some limit, or be the same order of magnitude as b . For example, a shape with area A and width w might have $A \sim w^2$. This sign is pretty loosely defined and usually means whoever is using it is “handwaving.” a and b usually the same units.

$a \lesssim b$: a is at most the order of magnitude of b . Yes, this means a can sometimes be greater than b . This isn’t used very much (thankfully). a and b have the same units.

$a \ll b$ or $a \lll b$, $b \gg a$: a is a *lot* smaller than b . The meaning of this is context dependent but usually $a/b \approx 0$. a and b have the same units (This is important! You can’t compare two things if they aren’t on the same scale).

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Reflection

Take some time to think through these questions. Part of purposeful practice is thinking through what your practice has taught you, how you will use it in the future, and what you might do differently next time, so we want to give you the space to do this.

1. What did you learn from these problems? How did you know to use the techniques you used? How might you apply this to a future problem?
2. How might you connect this to past material you've learned?
3. What questions do you have?